# Logistic regression:

⇒ Supervised learning alg.
⇒ classification prob.
⇒ predicts probabilities based on binary or multi-class outcomes using a logistic func (sigmoid func).

1) used for classification tasks:

↳ unlike linear regression (which predicts continuous vals), LogR is used for classification

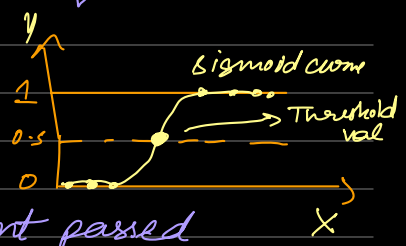
↳ predicts binary (yes/no/0/1) or multiclass (cat/dog/fish) labels.

↳ ex: Spam detection (spam/notspam)
disease diagnosis (+ve/-ve).

2) uses sigmoid func:

↳ o/p is a prob val (0,1), mapped using sigmoid func.

$$S(x) = \frac{1}{1+e^{-x}}$$




if $S(x) > 0.5$ ⇒ classify as 1 ex: Student passed
$S(x) < 0.5$ ⇒ classify as 0. ex: student failed.

3) linear decision boundary:

↳ decision bound is linear in simple log^c reg.
↳ despite the non-linear sigmoid func.

4) works for binary & Multi-class classification:

↳ Binary: classifies into two categories. ex: Spam/ not spam
Multiclass: classifies into 3 or more ~ ex: (apple, banana, orange)

5) Can handle continuous, categorical features:

↳ works with numerical, cat^n data
↳ One-Hot Encoding is used for categ^n data.

ex: predict loan approval based on income (numeric) & emp type (categorical)
- longest

6) feature scaling for best performance:

- features shd be standardized (Z-score / Min-Max Scaling).
- improves model accuracy.

7) robust to outliers:

- due to sigmoid func, extreme vals in data do not drastically affect predictions.
- uses regularization to reduce overfitting.

## Logistic Regression – Solved Example 1

1. Calculate the probability of pass for the student who studied 33 hours.

| Hours Study | Pass (1) / Fail (0) |
|---|---|
| 29 | 0 |
| 15 | 0 |
| 33 | 1 |
| 28 | 1 |
| 39 | 1 |

$s(x) = \frac{1}{1+e^{-x}}$

- $p = \frac{1}{1+e^{-z}}$
- $z = -64 + 2 * 33 = -64 + 66 = 2$
- $p = \frac{1}{1+e^{-2}} = 0.88$
- That is, if student studies 33 hours, then there is **88% chance** that the student will pass the exam

$\log(odds) = z = -64 + 2 * hours$

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## Logistic Regression – Solved Example 1

2. At least how many hours student should study that makes he will pass the course with the probability of more than 95%.

| Hours Study | Pass (1) / Fail (0) |
|---|---|
| 29 | 0 |
| 15 | 0 |
| 33 | 1 |
| 28 | 1 |
| 39 | 1 |

- $p = \frac{1}{1+e^{-z}} = 0.95$
- $0.95 * (1 + e^{-z}) = 1$
- $0.95 * e^{-z} = 1 - 0.95$
- $e^{-z} = \frac{0.05}{0.95} = 0.0526$
- [redacted]

$\ln(e^x) = x$

$-z = ln(0.0526) = -2.94$

$z = 2.94$

## Logistic Regression – Solved Example 1

| Hours Study | Pass (1) / Fail (0) |
|---|---|
| 29 | 0 |
| 15 | 0 |
| 33 | 1 |
| 28 | 1 |
| 39 | 1 |

- $z = 2.94$
- $\log(odds) = z = -64 + 2 * hours$
- $2.94 = -64 + 2 * hours$
- $2 * hours = 2.94 + 64$
- $2 * hours = 66.94$
- $hours = \frac{66.94}{2}$
- $hours = 33.47\ Hours$

- The student should study **at least 33.47 hours**, so that he will pass the exam with more than 95% probability